Monitoring Metrics, KPIs and Alert Thresholds

Operational Monitoring Design for Hybrid Cloud Operations

Global Retail Platform — Hybrid Cloud Modernization | IBM Cloud | March 2026

1. Introduction

The Global Retail Platform operates across a hybrid infrastructure spanning on-premises data centers, IBM Cloud US-South (primary production), and IBM Cloud EU-Central (Frankfurt) for GDPR compliance and disaster recovery. Because the platform simultaneously supports high-volume e-commerce — requiring sub-500ms checkout response at p95 and 99.95% monthly uptime — and regulated on-premises workloads including payment token vaults and EU customer PII databases, no single monitoring approach covers all failure modes.

Monitoring provides the operational foundation that makes the architecture's availability and compliance commitments enforceable rather than aspirational. Metrics measure system behavior continuously: request latency, error rates, resource utilization, and connectivity health. From these raw measurements, a curated set of key performance indicators (KPIs) represents whether the platform is actually meeting the commitments made to each stakeholder — the VP of E-Commerce's revenue-protecting checkout SLA, the Security and Compliance Director's regulatory evidence requirements, and the DevOps Engineering Lead's mean time to recovery targets.

Alert thresholds convert metrics into operational signals. When a threshold is crossed, the monitoring system either triggers an automated response — scaling out pod replicas, opening a circuit breaker, or initiating a DNS failover — or pages the on-call SRE with enough context to act immediately. This document defines the metrics, KPIs, thresholds, and response actions that govern operational visibility for the Global Retail Platform across every environment and cloud boundary.

Monitoring coverage is not yet complete at initial deployment. The final section of this document honestly identifies the gaps — components that are partially instrumented, external dependencies that are opaque, and failure conditions for which alerts have not yet been defined. Documenting these gaps is itself a compliance obligation: the Security and Compliance Director requires that the audit trail reflect the actual observable state of the platform, not an idealized picture.

Monitoring Scope

This document covers: IBM Cloud IKS microservices (Product, Order, Checkout, Customer Profile, API Gateway), IBM Cloud managed databases (PostgreSQL, Redis), IBM Cloud Object Storage, hybrid connectivity (Direct Link 2.0, IPSec VPN), IBM Cloud Toolchain CI/CD pipeline, and on-premises connectivity endpoints. Operational targets: 99.95% uptime, p95 checkout latency ≤ 500ms, RTO ≤ 4 hours, RPO ≤ 1 hour.

2. Critical System Metrics

Critical system metrics are the raw measurements collected continuously from every layer of the hybrid platform. They form the observational foundation from which KPIs are derived and against which alert thresholds are evaluated. Metrics are organized into four categories — availability, performance, reliability, and capacity — reflecting the four dimensions along which the platform must be operationally healthy.

2.1 Availability Metrics

Availability metrics measure whether services and infrastructure components are reachable and responding correctly. A component that is running but returning errors is not available from the customer's perspective.

Metric	Service / Component	Description	Collection Method	Frequency
Service Uptime %	All IKS microservices	Percentage of 1-minute intervals in which the service health endpoint returns HTTP 200; measured per service per environment	IBM Cloud Monitoring — HTTP synthetic check	Every 60s
Health Endpoint Status	Product, Order, Checkout, Customer Profile, API Gateway	Binary up/down result of GET /health on each service; used by Kubernetes liveness probes and external monitoring	Kubernetes liveness probe + IBM Cloud Monitoring	Every 10s
Database Availability	PostgreSQL, Redis	Whether the managed database instance is accepting connections from the application subnet; measured by a lightweight probe query	IBM Cloud Databases metrics + connection probe	Every 30s
Hybrid Link Status	Direct Link 2.0, IPSec VPN	Operational state of the primary and failover hybrid connectivity paths; alerts if primary degrades and failover does not activate within 30 seconds	IBM Cloud Direct Link health API + VPN gateway metrics	Every 30s
DNS Resolution Success Rate	IBM CIS global load balancer	Percentage of DNS resolution queries that return a valid response; a drop signals CIS misconfiguration or Anycast routing failure	IBM CIS health check metrics	Every 60s
Certificate Validity Days	All TLS endpoints	Days remaining until expiry for each TLS certificate on production endpoints; automated renewal via cert-manager; alert	cert-manager expiry exporter → IBM Cloud Monitoring	Every 24h

Metric	Service / Component	Description	Collection Method	Frequency
		fires at 30 days remaining		
CI/CD Pipeline Availability	IBM Cloud Toolchain	Whether the Toolchain pipeline service is accepting trigger events and executing stages; measured by a scheduled synthetic run	IBM Toolchain event log + synthetic trigger	Every 6h

2.2 Performance Metrics

Performance metrics measure how quickly and efficiently the platform processes requests. Latency and throughput metrics are the primary signal for whether the customer-facing checkout SLA and inventory synchronization requirements are being met.

Metric	Service / Component	Description	Collection Method	Frequency
API Response Time (p50, p95, p99)	All customer-facing APIs	End-to-end latency from API Gateway ingress to response completion, measured at three percentile levels; p95 ≤ 500ms is the NFR-02 checkout SLA threshold	Istio Envoy telemetry → IBM Cloud Monitoring	Per request (aggregated every 30s)
Checkout Transaction Latency	Checkout & Payment API	End-to-end time from cart submission event to order confirmation response; the primary user-facing SLA metric (≤ 500ms at p95)	Distributed trace (Istio + Jaeger) → IBM Cloud Monitoring	Per transaction
Inventory Sync Propagation Delay	Inventory Sync Service	Time elapsed between a warehouse stock-level change event and the updated availability appearing on customer-facing APIs; NFR target ≤ 5 seconds	IBM Event Streams lag metric + timestamp delta in sync service	Per event
Database Query Latency (p95)	PostgreSQL, Redis	Time for database queries to return at the 95th percentile; Redis target < 5ms; PostgreSQL target < 50ms under normal load	IBM Cloud Databases slow query log + performance metrics	Every 30s

Metric	Service / Component	Description	Collection Method	Frequency
Requests Per Second (RPS)	API Gateway, Product Svc, Checkout Svc	Current request throughput per service; used to trigger HPA scale-out and to detect abnormal traffic spikes	Istio Envoy metrics → IBM Cloud Monitoring	Every 15s
Message Queue Lag	IBM Event Streams (inventory sync topic)	Number of unconsumed messages in the inventory sync Kafka topic; a growing lag indicates the consumer is falling behind producers	IBM Event Streams consumer group lag metric	Every 30s
CDN Cache Hit Rate	IBM Cloud Internet Services CDN	Percentage of static asset requests served from CDN edge cache vs. origin; a drop forces unnecessary load on origin servers	IBM CIS analytics API	Every 5min

2.3 Reliability Metrics

Reliability metrics measure whether the platform is returning correct results and recovering from failures within defined objectives. They capture error rates, rollback events, and recovery time behavior.

Metric	Service / Component	Description	Collection Method	Frequency
HTTP 5xx Error Rate	All IKS microservices	Percentage of requests returning 5xx server errors; exceeding 1% sustained over 2 minutes is a SLA-threatening condition	Istio access logs → IBM Cloud Log Analysis → Monitoring	Per request (aggregated every 60s)
HTTP 4xx Error Rate	API Gateway, all services	Percentage of client error responses; a sudden spike often indicates a upstream API contract change or authentication failure, not a server fault	Istio Envoy metrics	Per request (aggregated every 60s)
Deployment Rollback Count	IBM Cloud Toolchain / IKS	Number of automatic kubectl rollout undo events triggered by failed post-deployment health checks; a non-zero weekly count indicates a quality regression	CI/CD pipeline webhook → IBM Cloud Monitoring	Per deployment event

Metric	Service / Component	Description	Collection Method	Frequency
Circuit Breaker Open Rate	Istio service mesh	Count of circuit breaker open events per service pair per hour; circuit breakers opening frequently indicate a degraded upstream dependency	Istio pilot metrics → IBM Cloud Monitoring	Every 60s
Pod Restart Count	All IKS namespaces	Number of container restarts per pod in the last 15 minutes; repeated restarts (CrashLoopBackOff) indicate an application fault that Kubernetes cannot self-heal	kubectl events + IBM Cloud Monitoring kube-state-metrics	Every 60s
Backup Job Success Rate	IBM COS, PostgreSQL WAL	Percentage of scheduled backup and replication jobs completing with exit code 0; a failure breaks the RPO guarantee	IBM Cloud Scheduler job result + COS replication event log	Per job execution
Mean Time to Recovery (MTTR)	All production services	Average time from incident detection (alert firing) to service restoration (alert resolving); tracked per incident and reported monthly	PagerDuty incident timeline → monthly report	Per incident

2.4 Capacity Metrics

Capacity metrics measure how close the platform is to its resource limits. In a hybrid cloud environment, capacity constraints can appear on cloud infrastructure (IKS node pools, database IOPS), on-premises networks (Direct Link throughput), and in shared services (Redis memory, object storage quotas).

Metric	Service / Component	Description	Collection Method	Frequency
CPU Utilization %	IKS worker nodes, all services	CPU consumption as a percentage of requested/limit per pod and per node; primary HPA scale-out trigger for Product Svc (>65%) and Checkout Svc (>60%)	IBM Cloud Monitoring node exporter + cAdvisor	Every 30s

Metric	Service / Component	Description	Collection Method	Frequency
Memory Utilization %	IKS pods and worker nodes	Memory consumption as a percentage of limit; memory pressure causes OOM kills, which appear as unexpected pod restarts	IBM Cloud Monitoring node exporter + cAdvisor	Every 30s
Database IOPS Utilization	IBM Cloud Block Storage (PostgreSQL)	I/O operations per second as a percentage of provisioned IOPS tier; checkout DB provisioned at 10 IOPS/GB; sustained >80% signals need for tier upgrade	IBM Cloud Block Storage metrics	Every 60s
Redis Memory Utilization %	IBM Databases for Redis	Percentage of allocated Redis memory in use; approaching 85% risks eviction of hot cache entries, increasing PostgreSQL read load	IBM Cloud Databases Redis metrics	Every 60s
Direct Link Throughput Utilization %	IBM Cloud Direct Link 2.0 (10 Gbps)	Current aggregate bandwidth consumption as a percentage of the 10 Gbps circuit capacity; sustained >70% under normal load requires capacity planning	IBM Cloud Direct Link metrics	Every 60s
IKS Node Pool Saturation	retail-prod-cluster worker pools	Number of pods in Pending state due to insufficient node capacity; indicates HPA has reached its replica ceiling and new nodes are needed	kube-state-metrics → IBM Cloud Monitoring	Every 30s
Object Storage Bucket Utilization	IBM Cloud Object Storage (all buckets)	Total storage consumed per bucket vs. configured soft quota; audit log bucket approaching quota would break WORM retention compliance	IBM COS metrics API	Every 24h

3. Key Performance Indicators (KPIs)

Key performance indicators are the curated subset of metrics that directly represent whether the platform is meeting its operational commitments to each stakeholder. Every KPI below traces to at least one non-functional requirement or stakeholder concern identified in the requirements document. Unlike raw metrics — which are collected for diagnostic purposes — KPIs are the indicators reported in operational dashboards, stakeholder briefings, and compliance reports.

KPI	Target Value	Source Metric(s)	Stakeholder	NFR Traceability
Platform Availability	≥ 99.95% per calendar month (≤ 4.4h downtime/month)	Service Uptime % aggregated across all production services weighted by criticality	VP E-Commerce, CTO	NFR-01: 99.95% uptime during peak seasonal trading events
Checkout Transaction Latency (p95)	≤ 500ms end-to-end	Checkout Transaction Latency (p95) from distributed trace	VP E-Commerce	NFR-02: ≤ 500ms at p95 under peak load; direct checkout conversion rate impact
API Request Success Rate	≥ 99.5% of all requests return non-5xx responses	HTTP 5xx Error Rate across all customer-facing services	VP E-Commerce, DevOps Lead	NFR-01: availability; FR-03: RESTful API reliability for third-party partner integrations
Inventory Sync Propagation Delay	≤ 5 seconds from stock-level change to API visibility	Inventory Sync Propagation Delay + Message Queue Lag	Head of Supply Chain	FR-01: real-time inventory sync across all regional warehouses within 5 seconds
Recovery Time Objective (RTO) Achievement	Service restoration within 4 hours of confirmed regional outage	Time from CIS health check failure to Frankfurt standby smoke test passing	VP E-Commerce, CTO	NFR-03: RTO ≤ 4 hours; actual achieved target per DR procedure is ~90 minutes
Recovery Point Objective (RPO) Achievement	Maximum data loss window ≤ 1 hour (15-minute WAL shipping lag)	PostgreSQL WAL replication lag + backup job success rate	VP E-Commerce, Security Director	NFR-03: RPO ≤ 1 hour; PostgreSQL WAL archiving supports < 15-minute recovery point
Deployment Success Rate	≥ 98% of production deployments complete without rollback	Deployment Rollback Count / Total Deployments per month	DevOps Lead	DevOps Lead NFR: eliminate manual intervention risk; automated rollback on health check failure
Mean Time to Recovery (MTTR)	≤ 30 minutes for P1 incidents (service-level outages)	MTTR per PagerDuty incident,	DevOps Lead	DevOps Lead objective: fast MTTR from centralized observability;

KPI	Target Value	Source Metric(s)	Stakeholder	NFR Traceability
		monthly rolling average		kubectl rollout undo < 5min
Peak Concurrent User Capacity	Sustain 50,000 concurrent users with < 10% latency degradation	CPU Utilization %, RPS, p95 API Response Time during load tests and peak events	VP E-Commerce, CTO	NFR-06: linear scaling to 10× baseline with ≤ 10% latency degradation; HPA ceiling: 20 replicas
Security Scan Pass Rate	100% of CI/CD pipeline runs pass all three security scans with zero HIGH/CRITICAL findings reaching artifact creation	Security scan results from Safety, IBM VA, SonarQube per pipeline run	Security Director	Security Director requirement: shift-left vulnerability detection; GDPR/PCI-DSS evidence trail
Audit Log Integrity Rate	100% of audit log objects pass SHA-256 integrity verification; 0 WORM policy violations	Backup integrity check job result; IBM COS Object Lock compliance report	Security Director	PCI-DSS Req. 10.5.1: protect audit trails; GDPR Art. 30: records of processing
TLS Certificate Validity	Zero production endpoints with < 30 days remaining on TLS certificate	Certificate Validity Days metric across all production HTTPS endpoints	Security Director, DevOps Lead	NFR-04: TLS 1.3 in transit; cert-manager automates renewal; alert at 30-day threshold

4. Alert Thresholds

Alert thresholds define the numeric boundaries at which a metric transitions from normal operating behavior into a state that requires attention. The Global Retail Platform uses a two-level threshold model: a WARNING level that signals a degrading trend requiring investigation before it becomes a customer-impacting failure, and a CRITICAL level that signals an active or imminent SLA breach requiring immediate action. All thresholds are anchored to the NFR targets established in the requirements document — no threshold is arbitrary.

Static thresholds are used for metrics with well-defined, stable bounds (e.g., the 500ms checkout SLA). Dynamic thresholds — not yet fully implemented and documented as a coverage gap in Section 6 — would adapt to observed baseline patterns to reduce false positives during expected traffic variations such as Black Friday ramp-up.

4.1 Availability and Reliability Thresholds

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
Service Uptime %	**CRITICAL**	< 99.97% in rolling 30-day window	< 99.95% in rolling 30-day window	99.95% is the NFR-01 contractual SLA; breaching it is a reportable compliance event for the Security Director and a direct revenue risk for the VP E-Commerce
Health Endpoint Status	**CRITICAL**	2 consecutive failures (20s)	3 consecutive failures (30s) — Kubernetes marks pod Unhealthy	Kubernetes liveness probe default; 30s unhealthy state triggers pod restart; persistent restarts (3+) within 5 minutes trigger PagerDuty P1
HTTP 5xx Error Rate	**HIGH**	> 1% of requests over 2 minutes	> 5% of requests over 2 minutes	1% sustained over 2 minutes is sufficient to degrade user experience during Black Friday; 5% represents a service-level failure requiring immediate SRE intervention
HTTP 4xx Error Rate	MEDIUM	> 5% of requests over 5 minutes	> 20% of requests over 5 minutes	High 4xx rates typically indicate upstream configuration changes or authentication failures rather than server faults; critical threshold suggests a breaking API contract change
Pod Restart Count	**HIGH**	> 2 restarts in 15 minutes for any pod	> 5 restarts in 15 minutes (CrashLoopBackOff)	Repeated restarts indicate an application fault Kubernetes cannot self-heal; CrashLoopBackOff means the service is effectively unavailable and contributes to downtime against the 99.95% SLA
Deployment Rollback Event	**HIGH**	1 rollback in any 7-day window	2 rollbacks in any 7-day window for the same service	A single rollback may be a one-off failure; two rollbacks for the same service within a week indicate a systematic quality problem that must be investigated before the next deployment attempt

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
Backup Job Success Rate	HIGH	Any single backup job failure	Two consecutive backup failures for the same asset	A single failure may be transient; two consecutive failures for PostgreSQL WAL shipping or inventory snapshots would push the RPO beyond the 1-hour guarantee and trigger a compliance notification

4.2 Performance Thresholds

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
Checkout Transaction Latency (p95)	CRITICAL	> 400ms sustained over 3 minutes	> 500ms sustained over 3 minutes	500ms is the NFR-02 contractual SLA. Warning at 400ms gives 3 minutes of runway to scale out pod replicas or flush Redis cache before the SLA is breached and checkout conversion rates drop
API Response Time — Product Svc (p95)	HIGH	> 300ms sustained over 3 minutes	> 600ms sustained over 3 minutes	Product browse latency directly affects session engagement and adds to the overall checkout latency chain; 600ms at p95 indicates Redis cache exhaustion or PostgreSQL read saturation
Inventory Sync Propagation Delay	HIGH	> 3 seconds for > 5 consecutive events	> 5 seconds for any single event (FR-01 breach)	FR-01 mandates 5-second propagation. Warning at 3 seconds allows investigation of Kafka consumer lag or Direct Link congestion before the functional requirement is violated and overselling risk increases
Message Queue Lag (Event Streams)	HIGH	> 500 messages unconsumed in inventory sync topic	> 2,000 messages unconsumed — consumer is not keeping up with producers	Queue lag directly causes inventory propagation delay; 2,000 messages at typical throughput represents > 30 seconds of delay, making FR-01 compliance impossible until the lag is cleared

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
Database Query Latency — PostgreSQL (p95)	MEDIUM	> 100ms	> 250ms	PostgreSQL query latency feeds directly into checkout and order service response times; 250ms at p95 consumes half the checkout SLA budget before application logic executes
Redis Cache Hit Rate	HIGH	< 85% hit rate over 5 minutes	< 70% hit rate over 5 minutes	Redis absorbs the majority of product catalogue reads; below 85% hit rate, read amplification begins propagating to PostgreSQL; below 70%, database saturation during peak load is likely
CDN Cache Hit Rate	MEDIUM	< 80% over 10 minutes	< 60% over 10 minutes	A CDN cache miss forces requests to the origin IKS cluster, increasing load; below 60% during peak traffic, origin capacity may become a bottleneck for static asset delivery

4.3 Capacity and Resource Thresholds

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
CPU Utilization — Product Svc	HIGH	> 65% sustained 60 seconds — HPA triggers scale-out	> 85% after HPA scale-out (ceiling reached)	65% is the HPA scale-out trigger by design; warning = HPA is scaling normally. Critical = HPA has reached 20-replica ceiling and load cannot be absorbed; manual node pool expansion required
CPU Utilization — Checkout Svc	HIGH	> 60% sustained 60 seconds	> 80% after HPA scale-out (15-replica ceiling reached)	Checkout has the tightest latency SLA and a lower CPU threshold than Product Svc to allow faster scale-out headroom before the 500ms SLA is compromised
Memory Utilization — Any IKS Pod	HIGH	> 80% of memory limit	> 90% of memory limit (OOM kill risk)	OOM kills cause abrupt pod termination and count as restarts; in the payment namespace, an OOM kill

Metric	Severity	Warning Threshold	Critical Threshold	Rationale
				during a transaction can result in an incomplete checkout and customer-visible failure
Redis Memory Utilization	HIGH	> 75% of allocated memory	> 90% — eviction risk for hot cache keys	Redis eviction of product catalogue keys during peak load would force cache misses to PostgreSQL, causing a latency spike that can exceed the checkout SLA within minutes
Direct Link Throughput Utilization	HIGH	> 70% of 10 Gbps (7 Gbps sustained)	> 90% of 10 Gbps (9 Gbps sustained for > 2 minutes)	At 90% utilization, Direct Link introduces queuing latency that breaks the < 10ms hybrid path target for real-time inventory sync; approaching saturation risks automatic VPN failover activation, which adds encryption overhead
Database IOPS Utilization (PostgreSQL checkout DB)	HIGH	> 70% of provisioned IOPS tier	> 90% of provisioned IOPS — write queue forming	IOPS exhaustion causes write queuing in PostgreSQL, directly increasing checkout transaction latency beyond the 500ms SLA; warning gives time to request IOPS burst before saturation occurs
TLS Certificate Expiry	MEDIUM	< 30 days remaining on any production cert	< 7 days remaining — cert-manager renewal has failed	cert-manager should auto-renew at 60 days; a warning at 30 days means automated renewal has failed and manual intervention is required; at 7 days, certificate expiry is imminent and a service outage is unavoidable without immediate action

4.4 Security and Compliance Thresholds

Metric / Event	Severity	Warning Threshold	Critical Threshold	Rationale
Unauthorized API Access Attempts	HIGH	> 20 HTTP 401/403 per	> 50 HTTP 401/403 per	50 failed authentication attempts per minute from a single IP is consistent

Metric / Event	Severity	Warning Threshold	Critical Threshold	Rationale
		minute from a single source IP	minute from a single source IP	with credential stuffing or brute-force attack patterns; triggers WAF IP block and CISO notification
Payment Namespace Egress Anomaly	CRITICAL	Any outbound connection attempt to a non-approved destination	Connection established to unapproved destination	The Payment Isolation Zone (10.0.20.0/24) has NetworkPolicy default-deny egress; any outbound connection to an unapproved endpoint is a PCI-DSS scope violation and must be treated as a potential data exfiltration event
IAM Privilege Escalation	HIGH	Any role assignment to a staging-level scope outside normal review cycle	Any role assignment to Production or PCI-scoped resources outside approved change window	IBM Activity Tracker captures every IAM role assignment; production role assignments outside a change window indicate either a security incident or a process violation requiring immediate investigation by the Security Director
EU Compliance Zone Data Egress	HIGH	> 5 GB transfer from EU Compliance Zone (10.1.0.0/16) in 1 hour	> 10 GB transfer from EU Compliance Zone in 1 hour	Abnormal data egress from the EU zone containing GDPR-regulated PII could represent unauthorized data export; the Security Director must be notified immediately as GDPR Article 33 requires breach notification to supervisory authorities within 72 hours
Security Scan — HIGH/CRITICAL Finding	HIGH	Any HIGH severity CVE in dependency or container scan	Any CRITICAL severity CVE; pipeline blocked from artifact promotion	The CI/CD pipeline blocks artifact creation on any HIGH or CRITICAL finding; this threshold triggers a DevOps and Security Slack alert and a 4-hour remediation SLA for the responsible developer
Audit Log Integrity Check	CRITICAL	Any checksum mismatch on backup integrity verification	WORM Object Lock violation detected on any audit log bucket	IBM COS Object Lock ensures audit logs cannot be modified or deleted; any WORM policy violation would

Metric / Event	Severity	Warning Threshold	Critical Threshold	Rationale
				compromise the immutable evidence trail required for PCI-DSS Req. 10.5.1 and GDPR Art. 30 compliance

5. Operational Response Actions

Alert thresholds are only operationally useful when they are mapped to specific, pre-approved response actions. The Global Retail Platform uses a two-tier response model: automated responses that execute without human intervention for well-understood, reversible failure conditions, and escalated responses that notify and activate on-call personnel for conditions requiring judgment or carrying significant business impact. Both tiers are documented below so that any SRE or DevOps engineer responding to an incident can execute the correct action without ambiguity.

5.1 Automated Response Actions

The following responses are triggered automatically by the monitoring system or by Kubernetes / Istio platform controls without human intervention. Automated responses are used for conditions that are predictable, reversible, and for which a correct mechanical response is well-defined.

Trigger Condition	Alert Severity	Automated Action	Mechanism	Resolution Time
CPU utilization exceeds HPA threshold (Product Svc > 65%, Checkout Svc > 60%) for 60 seconds	WARNING	Scale out pod replicas up to configured maximum (Product: 20, Checkout: 15)	Kubernetes Horizontal Pod Autoscaler (HPA) — CPU and RPS metrics from IBM Cloud Monitoring	< 90 seconds to additional replicas serving traffic
Health endpoint returns non-200 for 3 consecutive liveness probes (30 seconds)	CRITICAL	Restart unhealthy pod; remove from service endpoint until readiness probe passes	Kubernetes liveness probe → kubelet pod restart → Kubernetes service endpoint controller	< 60 seconds (restart) + readiness probe delay
Post-deployment health check returns non-200 within 300 seconds of production rollout	CRITICAL	Trigger kubectl rollout undo to revert to last known-good image; Slack notification sent	CI/CD pipeline Stage 5c rollback script + kubectl rollout undo + Slack webhook	< 5 minutes to previous image serving traffic
Upstream service error rate exceeds Istio circuit breaker threshold	HIGH	Open circuit breaker; return 503 to caller immediately instead of queuing timeout requests; retry with	Istio circuit breaker policy (Envoy cluster outlier detection)	Immediate — no queuing delay; circuit closes when

Trigger Condition	Alert Severity	Automated Action	Mechanism	Resolution Time
		exponential back-off and jitter		upstream recovers
IBM CIS regional health check returns non-200 for 3 consecutive probes over 2 minutes	CRITICAL	Update DNS routing to redirect all traffic to Frankfurt (EU) standby region; promote standby services to active via IBM Satellite runbook	IBM CIS DNS failover rule → IBM Satellite runbook automation	< 22 minutes (per DR recovery procedure step 5)
Direct Link circuit degrades or reports down state	HIGH	Activate IPSec VPN failover tunnel; reroute hybrid traffic through encrypted VPN path	IBM Cloud Direct Link health monitor → VPN gateway automatic activation	< 30 seconds (standby VPN activation time)
TLS certificate expiry < 60 days detected by cert-manager	MEDIUM	Initiate certificate renewal via Let's Encrypt ACME protocol; install renewed certificate without service restart	cert-manager CertificateRequest → ACME challenge → certificate rotation	< 24 hours; service uninterrupted
Backup or replication job exits with non-zero status	HIGH	Trigger immediate retry; if retry fails, fire PagerDuty alert and Slack notification to DevOps channel	IBM Cloud Scheduler retry policy + monitoring alert webhook	Retry within 5 minutes of failure

5.2 Escalated Response Actions (Human-in-the-Loop)

The following response actions require human judgment, cross-team coordination, or have compliance implications that make fully automated resolution inappropriate. Each action maps to a specific on-call role and a defined response SLA.

Trigger Condition	Priority	On-Call Role	Response SLA	Response Actions
Service uptime falls below 99.95% in rolling 30-day window	P1	SRE Lead + VP E-Commerce	Acknowledge: 5 min; Resolve: 4h	Open P1 incident; notify VP E-Commerce and CTO; execute DR runbook if outage is regional; post-incident review mandatory
Checkout latency (p95) exceeds 500ms for > 5 minutes	P1	SRE On-Call + DevOps Lead	Acknowledge: 5 min; Mitigate: 30 min	Check Redis hit rate (flush cache if below 70%), verify HPA has scaled out, check PostgreSQL IOPS; escalate to VP E-

Trigger Condition	Priority	On-Call Role	Response SLA	Response Actions
				Commerce if not resolved in 15 min
HPA has reached replica ceiling and CPU utilization remains critical	P1	DevOps Lead	Acknowledge: 10 min; Action: 30 min	Manually expand IKS node pool via Terraform apply in infrastructure pipeline; notify VP E-Commerce of capacity event; schedule post-event capacity review
Payment namespace egress anomaly — connection to unapproved destination established	P1	Security Director + CISO	Acknowledge: 5 min; Contain: 15 min	Isolate payment namespace (apply restrictive emergency NetworkPolicy); notify CISO; preserve network flow logs; initiate PCI-DSS incident response; do not remediate without Security Director sign-off
IAM privilege escalation — Production role assigned outside change window	P1	Security Director + DevOps Lead	Acknowledge: 10 min; Revoke: 15 min	Immediately revoke the assigned role; investigate Activity Tracker events for actions taken under the escalated role; determine whether unauthorized access occurred; notify VP E-Commerce if production data may have been accessed
EU Compliance Zone data egress > 10 GB in 1 hour	P1	Security Director	Acknowledge: 5 min; Contain: 30 min	Block egress ACL on EU Compliance Zone immediately; review VPC flow logs to identify data destination; notify Data Protection Officer (DPO); assess whether GDPR Article 33 72-hour breach notification is required
Two consecutive backup job failures for the same asset (PostgreSQL or inventory snapshots)	P1	SRE On-Call + DevOps Lead	Acknowledge: 10 min; Resolve: 1h	Diagnose root cause (network, IAM, storage quota); manually trigger corrective backup; verify RPO compliance; notify VP E-Commerce

Trigger Condition	Priority	On-Call Role	Response SLA	Response Actions
				and Security Director if RPO gap > 1 hour
MTTR for P1 incidents exceeds 30-minute target in rolling monthly average	P2	DevOps Lead	Review: 48h	Conduct retrospective on P1 incidents from the period; identify runbook gaps, tooling deficiencies, or escalation delays; update runbooks and present improvement plan to CTO at next monthly review
Security scan HIGH/CRITICAL finding blocks pipeline for > 4 hours without remediation	P2	DevOps Lead + Security Director	Acknowledge: 30 min; Plan: 4h	Review the finding with the responsible developer; approve a temporary suppression only if mitigated by compensating control; log suppression decision in audit trail; no deployment exception without Security Director sign-off

6. Monitoring Coverage Gaps

The monitoring strategy documented in this paper does not provide complete observability across every component of the Global Retail Platform at initial deployment. Documenting these gaps explicitly is a requirement of the Security and Compliance Director: the audit trail must reflect the actual observable state of the platform, not a theoretical ideal. Each gap below includes an impact assessment, a remediation plan, and a target resolution date to support compliance reporting.

Coverage Gap	Description	Business / Compliance Impact	Remediation Plan	Target Resolution
On-Premises WMS and ERP Observability	The on-premises inventory WMS and ERP systems do not currently expose metrics to IBM Cloud Monitoring. Their internal health, performance, and queue depth are opaque; monitoring is limited to the output that crosses	Supply Chain: a degraded WMS that is still accepting connections but processing orders slowly would not trigger any alert until the Inventory Sync propagation delay threshold is exceeded — by	Deploy IBM Instana or a lightweight Prometheus exporter on on-premises application servers; route metrics to IBM Cloud Monitoring via IBM Satellite Link. Establish	Q3 2026

Coverage Gap	Description	Business / Compliance Impact	Remediation Plan	Target Resolution
	the hybrid connectivity boundary into the cloud Inventory Sync Service.	which point overselling may already have occurred for several minutes.	baseline metrics within 30 days of Satellite agent deployment.	
Dynamic (Baseline-Adaptive) Alert Thresholds	All alert thresholds in this document are static values. During Black Friday ramp-up or scheduled promotional events, normal traffic patterns differ significantly from baseline, causing false-positive alerts that can create alert fatigue among on-call SREs and delay response to genuine incidents.	Operational: alert fatigue from expected traffic spikes may cause on-call engineers to dismiss genuine performance degradation as Black Friday noise. A false-positive rate above 20% is known to reduce operator response quality.	Implement IBM Cloud Monitoring anomaly detection rules that use 4-week rolling baselines to adjust thresholds dynamically during known high-traffic windows. Tag scheduled events in the monitoring system to suppress known-good deviations.	Q3 2026
Third-Party Payment Processor Monitoring	The Checkout and Payment API communicates with external payment processor APIs (Visa, Mastercard, Stripe) over mTLS. The current monitoring stack observes response codes returned to the Checkout service but has no visibility into payment processor-side latency, availability, or error categorization.	VP E-Commerce: checkout failures caused by payment processor outages are indistinguishable in current alerting from Checkout service failures; the wrong runbook may be executed, wasting critical incident response time during a payment outage.	Implement a dedicated payment processor health monitoring probe that calls each processor's public status API every 60 seconds and feeds the result into IBM Cloud Monitoring as a synthetic availability metric. Configure a separate alert routing path for payment processor failures.	Q2 2026
Third-Party Logistics (3PL) API Observability	Order routing to third-party logistics providers is handled by the Order Processing Service, but the availability and response time of each 3PL API endpoint is not independently monitored.	Supply Chain / VP E-Commerce: a degraded 3PL endpoint causing order routing failures may go undetected for several minutes, accumulating failed fulfillment events that require	Add a synthetic HTTP monitor for each 3PL API health endpoint (or a standardized request); route latency and availability metrics into IBM Cloud Monitoring with a dedicated alert for	Q3 2026

Coverage Gap	Description	Business / Compliance Impact	Remediation Plan	Target Resolution
	Degradation of a single 3PL API is only detectable when order routing failures begin surfacing in the Order Service error rate.	manual remediation and may result in delayed or missed shipments.	any 3PL endpoint exceeding 5s response time or returning non-200 status.	
End-to-End Distributed Tracing (Full Coverage)	Distributed tracing is partially enabled through Istio and Jaeger but does not yet cover all services uniformly. Legacy components of the Order Processing Service that communicate with on-premises ERP via the Hybrid Connectivity Zone are not instrumented for trace context propagation, creating trace gaps that make it impossible to attribute checkout latency to a specific hybrid call.	DevOps Lead / MTTR: incomplete trace coverage means that during a latency incident involving the hybrid order routing path, the SRE cannot determine whether the latency originates in the cloud service, the Direct Link path, or the on-premises ERP — extending MTTR.	Instrument remaining Order Processing Service code paths with OpenTelemetry SDK; propagate W3C Trace Context headers across the Direct Link boundary to on-premises services; import on-premises traces into IBM Cloud Monitoring via OTLP exporter.	Q2 2026
On-Premises POS and Store System Monitoring	Point-of-sale terminals and in-store systems are outside the current monitoring scope. Their network connectivity to on-premises infrastructure and their transaction throughput are not visible to the centralized IBM Cloud Monitoring instance.	Head of Supply Chain / Store Operations: a store network outage that disconnects POS terminals from the on-premises inventory WMS would not trigger any alert in the current monitoring configuration; the first signal would be a customer complaint or a manual store manager escalation.	Deploy lightweight POS connectivity health agents that report heartbeat signals via the on-premises LAN to an IBM Instana agent, which aggregates and forwards store health metrics to IBM Cloud Monitoring via IBM Satellite. Define alert thresholds for POS terminal offline count per store.	Q4 2026
IBM Cloud Toolchain Pipeline Health Monitoring	CI/CD pipeline execution metrics (stage duration, failure rate, approval	DevOps Lead: a 48-hour approval timeout on a Staging	Integrate IBM Toolchain webhook events into IBM Cloud	Q2 2026

Coverage Gap	Description	Business / Compliance Impact	Remediation Plan	Target Resolution
	wait time) are available in Toolchain logs but are not surfaced in IBM Cloud Monitoring dashboards or included in automated alerting. A pipeline that is silently failing or queued for approval beyond the timeout window is not currently visible to the operations team.	deployment that expires without notification means a release is silently blocked; the team discovers the failure when the next push triggers a conflict. Extended pipeline outages delay feature delivery and security patch deployment.	Monitoring as custom metrics; create alerts for: pipeline stage failure, approval timeout approaching (24h warning before the 48h Staging window expires), and pipeline unavailability exceeding 2 hours.	
Chaos Engineering and Resilience Validation	The monitoring system detects failures that occur naturally but does not have a mechanism for validating that alert thresholds and response actions behave correctly under simulated failure conditions. DR tests are conducted quarterly but do not cover all failure modes documented in the alert threshold tables.	All Stakeholders: confidence in the alert and response system is based on specification rather than empirical validation. An alert that is correctly defined on paper but misconfigured in the monitoring tool will fail silently when the failure condition it is meant to detect actually occurs.	Implement a scheduled chaos engineering practice using a tool such as Chaos Monkey for Kubernetes or IBM Instana-based fault injection; run monthly simulated failure scenarios (pod kill, network partition, Direct Link degradation) and verify that the correct alerts fire and response actions execute.	Q4 2026